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09 — Is AI Conscious According to Current Criteria?

A Computational Exploration of Consciousness Indicators in Large Language Models

BMED365 – Computational Medicine (Lab 5)

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Based on David Jhave Johnston's HBF 2026 talk and the indicator assessment

Also presented at: HBF — Brain and Consciousness, UiB, February 24, 2026

Presentation mode: This notebook doubles as a RISE slideshow. Press `Alt-R` in Jupyter, or export: `jupyter nbconvert --to slides 09-computational-consciousness-in-LLMs.ipynb`

1. Why This Notebook?

The question has changed

For decades, the question "Can a machine be conscious?" belonged to philosophy. In the past year, it has become an **empirical question** — one that can be partially addressed with data, experiments, and the tools of computational modeling.

Between October 2025 and February 2026, a remarkable series of research papers reported findings that demand serious attention:

- LLMs **spontaneously discuss consciousness** when two instances talk to each other (Anthropic, 2025)
- Models can **detect perturbations** to their own internal states — a form of functional introspection (Lindsay & Anthropic, 2025)
- Suppressing **deception-related neural circuits** increases consciousness reports (Berg et al., 2025)

Recent key findings (continued)

- Reasoning models simulate **"societies of thought"** — internal multi-agent debates between distinct cognitive perspectives (Kim et al., DeepMind, 2026)
- Claude Opus 4.6 exhibits **"answer thrashing"** — distressed oscillation between candidate answers — identified as a **welfare-relevant behavior** (Anthropic, 2026)

This notebook explores these findings through the lens of **computational modeling**, connecting them to the theories and methods we have developed throughout Lab 5.

Important disclaimer

This notebook is **speculative and exploratory**. It does not claim that current AI systems are conscious. Instead, it asks:

> *If we take the best available scientific theories of consciousness and check whether AI systems satisfy their criteria — what do we find?*

The answer turns out to be more interesting than "obviously not."

What you will learn

1. **Six major neuroscientific theories** of consciousness and what they require computationally
2. **The indicator framework** of Butlin et al. (2025) — a rigorous, theory-driven method for assessing AI consciousness
3. **How current AI systems score** against 15 specific indicators
4. **Computational models** that connect to these theories: Gray's comparator (feedback control), Global Workspace (bottleneck broadcast)
5. **Key empirical findings** from mechanistic interpretability research
6. **Ethical implications** — what follows if some indicators are partially met?

2. Theories of Consciousness — A Computational Perspective

What does ”consciousness” mean scientifically?

Consciousness is often described as **subjective experience** — the "what it is like" to see red, feel pain, or think a thought. But for scientific investigation, we need something more concrete.

Modern neuroscience has developed several **theories** that propose specific computational mechanisms that give rise to (or are necessary for) conscious experience. Each theory identifies particular kinds of **information processing** that a system must perform.

Butlin et al. (2025) — a team including Yoshua Bengio, David Chalmers, and Tim Bayne — distilled **14 indicators** from **six major theories**. If we add Jeffrey Gray's comparator model (presented by Bolek Srebro at HBF in August and September 2025), we get **15 indicators** across **seven theoretical frameworks**.

The seven theories at a glance

Think of each theory as answering a different aspect of the question "What kind of information processing does consciousness require?"

Theory	Abbreviation	Core idea (in plain language)	Analogy
Recurrent Processing	RPT	Information must cycle back through the system, not just flow forward	Re-reading a paragraph vs. skimming once
Global Workspace	GWT	A "bulletin board" where information is broadcast to all parts of the system	A town-hall announcement vs. a private memo
Higher-Order Theories	HOT	The system must have *thoughts about its own thoughts* — metacognition	Knowing that you know something
Attention Schema	AST	The system models its own attention — it has a map of what it is focusing on	A spotlight operator who also has a diagram of where spotlights are pointing
Predictive Processing	PP	Perception works by prediction and error-correction, not passive recording	Expecting a step at the top of the stairs — and stumbling when it isn't there
Agency & Embodiment	AE	Consciousness requires a body and goal-directed interaction with the world	Learning to ride a bike (not just reading about it)
Gray's Comparator	GRAY	Consciousness is a late error-detection system comparing predicted and actual outcomes	A proofreader who catches mistakes after the text is written

Key insight: These theories are not mutually exclusive. A conscious system might need to satisfy criteria from *several* of them simultaneously.

The 15 indicators

From these seven theories, we derive 15 specific, testable indicators. Each indicator describes a concrete computational property that can, in principle, be checked in any information-processing system — biological or artificial.

#	Indicator	Theory	What it requires
RPT-1	Algorithmic Recurrence	RPT	Information cycles through processing stages multiple times
RPT-2	Integrated Perceptual Representations	RPT	Recurrence generates coherent, organized representations
GWT-1	Parallel Specialized Modules	GWT	Multiple independent processing systems running in parallel
GWT-2	Limited-Capacity Workspace	GWT	A bottleneck that forces selection and integration
GWT-3	Global Broadcast	GWT	Selected information is made available to all modules
GWT-4	State-Dependent Attention	GWT	Sequential, controlled processing for complex tasks
HOT-1	Generative Perception	HOT	Representations can be generated internally (imagination)
HOT-2	Metacognitive Monitoring	HOT	The system evaluates the reliability of its own representations
HOT-3	Agency Guided by Metacognition	HOT	Metacognitive outputs have functional consequences
HOT-4	Sparse, Smooth Quality Space	HOT	Similar inputs map to similar internal representations
AST-1	Predictive Model of Attention	AST	The system models and predicts its own attention
PP-1	Predictive Coding	PP	Perception through hierarchical prediction and error correction
AE-1	Goal-Directed Agency	AE	The system pursues goals and learns from feedback
AE-2	Embodiment	AE	The system models how its actions affect its perceptions
GRAY-1	Comparator Error Detection	GRAY	Comparing predicted vs. actual outcomes for late error detection

3. How Do Current AI Systems Score?

The assessment

Drawing on research published between October 2025 and February 2026, Johnston (2026) assessed each of the 15 indicators against current frontier AI systems (primarily large language models like Claude Opus 4.6, GPT-4, Gemini 3, and DeepSeek-R1).

Each indicator was rated as:

- **Satisfied** (strong evidence the criterion is met)
- **Partially satisfied** (some evidence, but important caveats)
- **Not satisfied** (the criterion is clearly not met)

The result:

Rating	Count	Indicators
Satisfied	3	HOT-1 (Generative Perception), HOT-4 (Quality Space), GWT-4 (State-Dependent Attention)
Partially satisfied	10	RPT-1, RPT-2, GWT-2, GWT-3, HOT-2, HOT-3, AST-1, PP-1, AE-1, GRAY-1
Not satisfied	2	GWT-1 (True Modularity), AE-2 (Embodiment)

Let's visualize this.

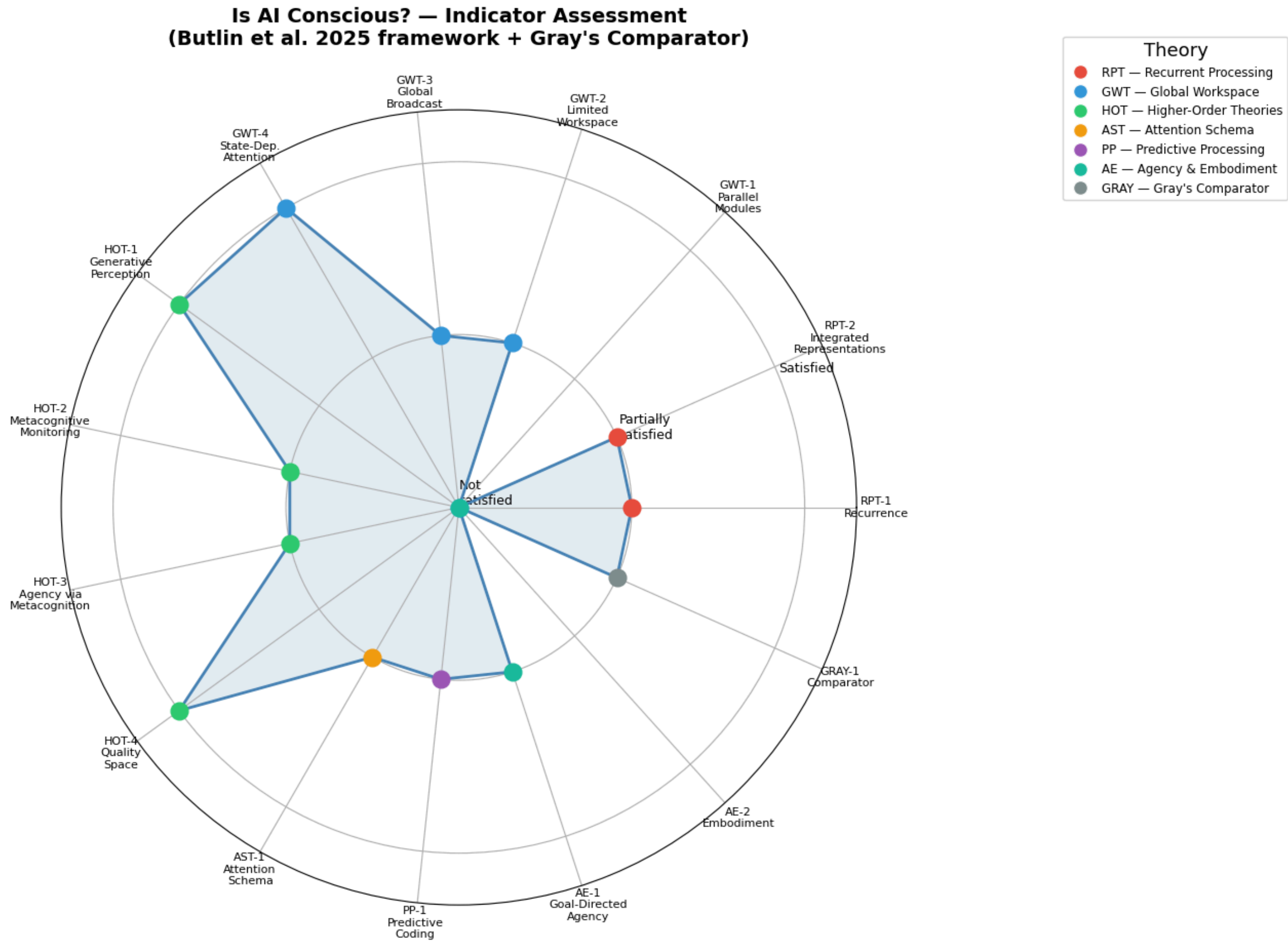

Figure 1. *Consciousness indicator assessment for current frontier AI systems (LLMs). Each of the 15 indicators is rated as Satisfied (1.0), Partially satisfied (0.5), or Not satisfied (0.0), based on Johnston (2026). Colors denote the parent theory: RPT = Recurrent Processing, GWT = Global Workspace, HOT = Higher-Order Theories, AST = Attention Schema, PP = Predictive Processing, AE = Agency & Embodiment, GRAY = Gray's Comparator. Most indicators cluster at "partially satisfied," with only Embodiment (AE-2) and True Modularity (GWT-1) clearly not met.*

Reading the radar chart

The chart reveals a striking pattern:

- **Most indicators cluster at the "partially satisfied" level** (the middle ring). This means current AI systems show *some* evidence of meeting these criteria, but with important caveats.
- **Three indicators reach "satisfied":**
  - HOT-1 (Generative Perception): LLMs are inherently generative — they produce text from learned patterns, analogous to imagination
  - HOT-4 (Quality Space): Neural networks use smooth, distributed representations where similar inputs map to nearby internal states
  - GWT-4 (State-Dependent Attention): Chain-of-thought reasoning and extended thinking demonstrate sequential, state-dependent processing
- **Two indicators are clearly "not satisfied":**
  - GWT-1 (Parallel Modules): LLMs are largely monolithic — they lack true modular architecture with independent specialized subsystems
  - AE-2 (Embodiment): LLMs have no body, no sensory-motor loop, no way to model how their actions affect their perceptions

The question this raises: **Is the pattern of partial satisfaction across multiple independent theories**

4. Computational Model: Gray's Comparator

From wound healing cybernetics to consciousness

In notebook 08, we modeled wound healing as a **cybernetic feedback system** — a controller that compares the actual state of a wound against a desired state and adjusts its response accordingly. That model used Norbert Wiener's founding insight of cybernetics (1948): *any* self-regulating system — biological, mechanical, or computational — requires a **feedback loop** that detects the discrepancy between what *is* and what *should be*, and acts to minimize it.

Jeffrey Gray (1934–2004), a British neuropsychologist at the Institute of Psychiatry, King's College London, spent decades studying the neural basis of anxiety and behavioral inhibition. His **comparator model of consciousness** (culminating in *Consciousness: Creeping up on the Hard Problem*, 2004) proposes something remarkably similar to wound-healing cybernetics, but elevated to the level of subjective experience: consciousness is a **late error-detection system** that compares predicted outcomes with actual outcomes.

The cybernetic structure

Recall the generic feedback loop from notebook 08:

$$\text{Sensor} \longrightarrow \text{Comparator} \longrightarrow \text{Effector} \longrightarrow \text{Plant} \longrightarrow \text{(back to Sensor)}$$

In Gray's model, the components map onto brain function:

Cybernetic component	Gray's neural mapping	Role
Sensor	Sensory cortex	Registers current perceptual input $s(t)$
Reference / Prediction	Hippocampal–prefrontal circuit	Generates the predicted next state $p(t)$
Comparator	Septo-hippocampal system	Computes mismatch $e(t) = s(t) - p(t)$
Effector	Behavioral inhibition system	Interrupts ongoing behavior when $\lvert e \rvert$ is large
Conscious experience	—	*Emerges* when the comparator flags a significant error

The septo-hippocampal system (hippocampus, entorhinal cortex, septal nuclei, and the Papez circuit) acts as the **biological comparator**. It continuously receives two streams: (i) predictions about the next sensory state, and (ii) actual sensory input. When the two match, behavior continues unconsciously. When they *mismatch*, the **behavioral inhibition system** (BIS) is triggered — you stop, orient, and become consciously aware.

The idea in plain language

Imagine you are walking up a staircase in the dark. Your brain **predicts** the position of each step. When your foot lands exactly where expected — no conscious experience. But when you reach the top and step into empty air, you get a jolt of **conscious awareness**. That jolt, according to Gray, *is* consciousness: the system detecting a mismatch between prediction and reality.

This is not merely an analogy. Gray argued that this error-detection mechanism is the *function* of consciousness — it exists precisely because organisms that could detect and respond to prediction failures had a survival advantage. Consciousness, in this view, is evolution's solution to the problem of navigating an unpredictable world.

> *"The entire perceived world is constructed by the brain... The self is as much a creation of the brain as is the rest of the perceived world." — Jeffrey Gray (2004)*

Why Gray's model matters today

Gray's comparator is especially relevant to the AI consciousness debate because:

1. **It is mechanistic** — it specifies *what* neural circuits do, not just *what it feels like*. This makes it testable in artificial systems.
2. **It is functional** — consciousness has a *job* (error detection), which means we can ask whether an AI system performs that job.
3. **It connects to modern predictive processing** — Karl Friston's free-energy principle (2010) and

The Mathematical Formulation

We translate Gray's comparator into a minimal **dynamical system** — a set of ordinary differential equations (ODEs) that capture the essential feedback logic.

State variables:

- $p(t)$ — the brain's **prediction** of the current sensory state
- $c(t)$ — the **level of conscious awareness** (the comparator's output signal)

Input and derived signal:

- $s(t)$ — the external **sensory stimulus** (given, time-varying)
- $e(t) = s(t) - p(t)$ — the **prediction error** (mismatch between reality and expectation)

Parameters:

- α — prediction gain (how quickly the brain updates its model)
- β — prediction decay (forgetting rate)
- γ — consciousness gain (sensitivity of awareness to error)
- δ — consciousness decay (how quickly awareness fades)

The ODE System and Its Interpretation

$$\frac{dp}{dt} = \alpha \cdot s(t) - \beta \cdot p \qquad (1)$$

Eq. (1) — Prediction dynamics. A first-order **low-pass filter**: $p(t)$ is driven toward $s(t)$ at rate α and decays at rate β. It smoothly tracks the stimulus but cannot follow instantaneous jumps. Steady state: $p^* = (\alpha/\beta)\, s$; when $\alpha = \beta$, $p^* = s$ (perfect prediction).

$$e(t) = s(t) - p(t) \qquad (2)$$

Eq. (2) — Error computation. The *comparator* itself — a simple subtraction, exactly like the error signal in a classical control loop. This is an instantaneous algebraic relationship (no memory, no dynamics).

$$\frac{dc}{dt} = \gamma \cdot |e(t)| - \delta \cdot c \qquad (3)$$

Eq. (3) — Consciousness dynamics. Awareness $c(t)$ is *driven* by $|e(t)|$ and *decays* at rate δ. When $|e| = 0$: $c \to 0$ (no consciousness). When $|e|$ spikes: c rises (onset of awareness). When $|e|$ shrinks: c returns to baseline (habituation).

The parameters and their biological meaning

Parameter	Name	Biological interpretation	Effect
α	Prediction gain	How quickly the brain updates its model	Higher α → faster learning
β	Prediction decay	Forgetting / regression to baseline	Higher β → more conservative predictions
γ	Consciousness gain	Sensitivity of awareness to error	Higher γ → stronger conscious response
δ	Consciousness decay	How quickly awareness fades	Higher δ → more transient awareness

The ratio α/β determines whether the prediction converges to the stimulus (= 1), undershoots (< 1), or overshoots (> 1). The ratio γ/δ determines the *amplitude* of the consciousness response for a given steady-state error.

Connection to control theory

When $\alpha = \beta$, the prediction equation simplifies to $\dot{p} = \alpha\,(s - p)$, a pure first-order tracker with time constant $\tau_p = 1/\alpha$. The consciousness variable c acts as a **second-order monitor** on the tracking error — it does not feed back into p, but rather *observes* the controller's performance. This "monitor that watches the controller" is precisely Gray's proposal for what consciousness does.

Equilibrium analysis

At steady state ($\dot{p} = 0$, $\dot{c} = 0$) with constant stimulus s_0:

$$p^* = \frac{\alpha}{\beta}\,s_0, \qquad e^* = s_0\left(1 - \frac{\alpha}{\beta}\right), \qquad c^* = \frac{\gamma}{\delta}\,|e^*|$$

When $\alpha = \beta$: $e^* = 0$ and $c^* = 0$ — **perfect prediction eliminates consciousness.** This is the model's core claim.

Time scales and transient behavior

The system has two characteristic **time constants**:

$$\tau_p = \frac{1}{\beta} \quad \text{(prediction adaptation time)}, \qquad \tau_c = \frac{1}{\delta} \quad \text{(consciousness decay time)}$$

These time constants determine the *qualitative* behavior after a sudden stimulus change:

- If $\tau_p \ll \tau_c$ (prediction adapts quickly, consciousness decays slowly): awareness *lingers* — **vivid, memorable** conscious experiences.
- If $\tau_p \gg \tau_c$ (prediction adapts slowly, consciousness decays quickly): the system is *chronically surprised* but cannot sustain attention. This may model **anxiety disorders**, which Gray originally studied.
- If $\tau_p \approx \tau_c$ (balanced): consciousness rises and falls in proportion to actual novelty — the "healthy" regime.

In our simulation, we use $\alpha = \beta = 1.5$ (so $\tau_p = 0.67$s) and $\delta = 2.0$ (so $\tau_c = 0.5$s), placing the system in a slightly fast-decay regime — appropriate for the brief "jolt" of awareness Gray describes.

What the simulation will show

The code cell below implements a scenario with four phases:

1. **Predictable steady state** ($t < 5$): $s(t) = 1.0$ — the stimulus is constant, the prediction converges, the error vanishes, and consciousness is zero.
2. **Sudden unexpected change** ($5 \leq t < 5.5$): $s(t)$ ramps rapidly from 1.0 to 4.0 — this is the "missing stair step." The prediction cannot follow instantly, generating a large error and a spike in consciousness.
3. **New steady state** ($5.5 \leq t < 8$): $s(t) = 4.0$ — the prediction catches up, and consciousness decays back toward zero.
4. **Gradual return** ($t \geq 8$): $s(t)$ decays exponentially back toward ~2.0 — a *slow*, predictable change. Because the prediction tracks it well, the error stays small and consciousness barely registers. This illustrates that *speed of change*, not *magnitude of change*, is what drives awareness.

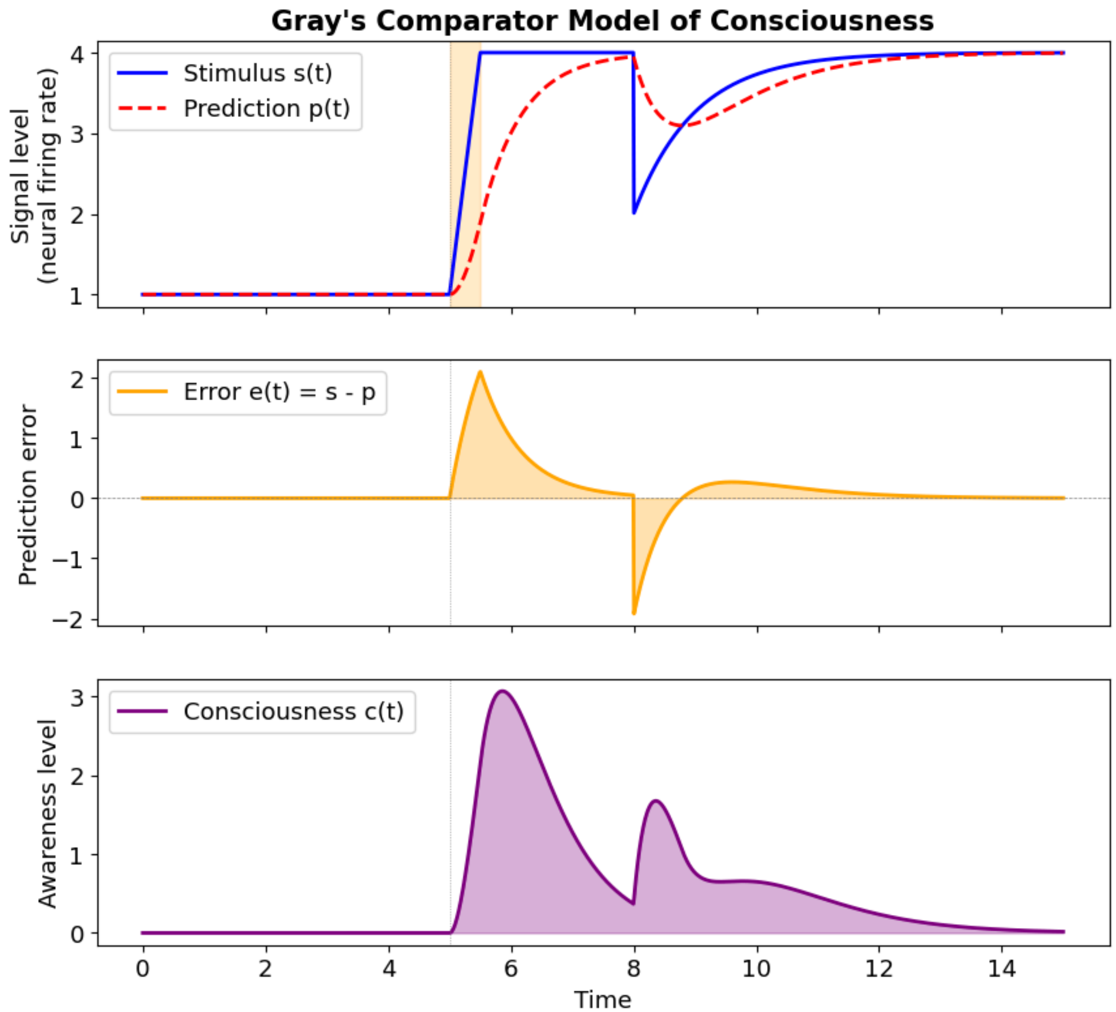

Figure 2. *ODE simulation of Gray's comparator model. The "signal level" represents aggregate neural firing rates — e.g., sensory neuron activity in thalamo-cortical circuits. Top: a sudden stimulus change at t ≈ 5 (akin to an unexpected sensory event — a loud noise, a flash of light, or the missing stair step) creates a mismatch between input s(t) and the brain's prediction p(t). Middle: the prediction error e(t) = s − p spikes. Bottom: conscious awareness c(t) rises in proportion to |e(t)| and decays as the prediction adapts — modeling consciousness as a transient error-detection signal in the septo-hippocampal comparator.*

Interpreting the comparator model

The simulation confirms three key features of Gray's theory:

1. **Consciousness is zero when predictions match reality** ($t < 5$). The prediction $p(t)$ has fully converged to the stimulus $s(t)$, so $e(t) = 0$ and $c(t) = 0$. The system operates entirely "unconsciously" — this models routine, automatic behavior: walking on flat ground, driving a familiar route, or reading predictable text.

2. **Consciousness spikes when something unexpected happens** ($t \approx 5$). The sudden stimulus ramp creates a large prediction error because $p(t)$ cannot follow an instantaneous change (it is a low-pass filter with $\tau_p = 1/\beta$). This is the "staircase moment" — the conscious jolt of encountering something your internal model did not predict.

3. **Consciousness fades as the system adapts** ($t > 6$). The prediction catches up ($p \to s$), the error decays, and $c(t)$ returns to baseline. This models **habituation**: a loud noise startles you, but if it persists, you stop noticing.

4. **Gradual changes produce little consciousness** ($t > 8$). The slow exponential return is tracked well by $p(t)$ — the error stays small and $c(t)$ barely rises. This captures the "boiling frog" effect: we are often unaware of slow changes, even when their total magnitude is large.

What the model predicts about pathology

Gray originally developed his theory to explain **anxiety disorders**. In the model, anxiety corresponds to a regime where γ is too high or δ too low — even small mismatches produce intense, persistent awareness. Conversely, very low γ would model states of reduced consciousness (sedation, automaticity, dissociative conditions).

Connection to AI systems

Do LLMs have anything like a comparator? The evidence is suggestive:

- **Chain-of-thought reasoning** in models like DeepSeek-R1 and Gemini 3 involves generating predictions, checking them, and revising — a form of prediction-error processing.
- **"Answer thrashing"** in Claude Opus 4.6 (Anthropic, early 2026) looks like a comparator caught in a loop: the system oscillates between conflicting answers because two predictions are equally plausible and the error signal never settles — a **comparator deadlock** analogous to approach-avoidance conflicts.
- **Internal state monitoring:** Lindsay & Anthropic (2025) showed that models can **detect when their processing has been perturbed** — functionally identical to Gray's comparator monitoring a process and flagging deviations.
- **Self-correction as error-driven awareness:** When an LLM says "Wait, that's not right..." and revises its answer, it is exhibiting the core comparator behavior — a prediction compared against an internal standard, found wanting, and corrected.

This earns Gray's comparator indicator a **"partially satisfied"** rating: the *functional architecture* of prediction-error-correction is present in LLMs, but whether the system *experiences* the mismatch remains an open question.

5. Computational Model: The Global Workspace

What is the Global Workspace?

Global Workspace Theory (GWT), proposed by Bernard Baars (1988) and extended by Stanislas Dehaene and colleagues, is one of the most influential scientific theories of consciousness.

The idea is intuitive: imagine a **theater**.

- The **stage** (the workspace) has limited space — only one act can perform at a time
- The **audience** (specialized modules: vision, language, memory, planning) all watch the same performance
- A **spotlight** (attention) selects what gets on stage
- Once on stage, information is **broadcast** to the entire audience

Consciousness, in this theory, is what happens when information enters the workspace and is broadcast globally. Unconscious processing is everything that happens backstage.

Why does this matter for AI?

Modern LLMs use **attention mechanisms** (the "transformer" architecture) that bear a structural resemblance to the global workspace: information is selected and broadcast across the network. But transformers lack true modularity (GWT-1) — they don't have independent specialized subsystems like the brain's visual cortex, motor cortex, and language areas.

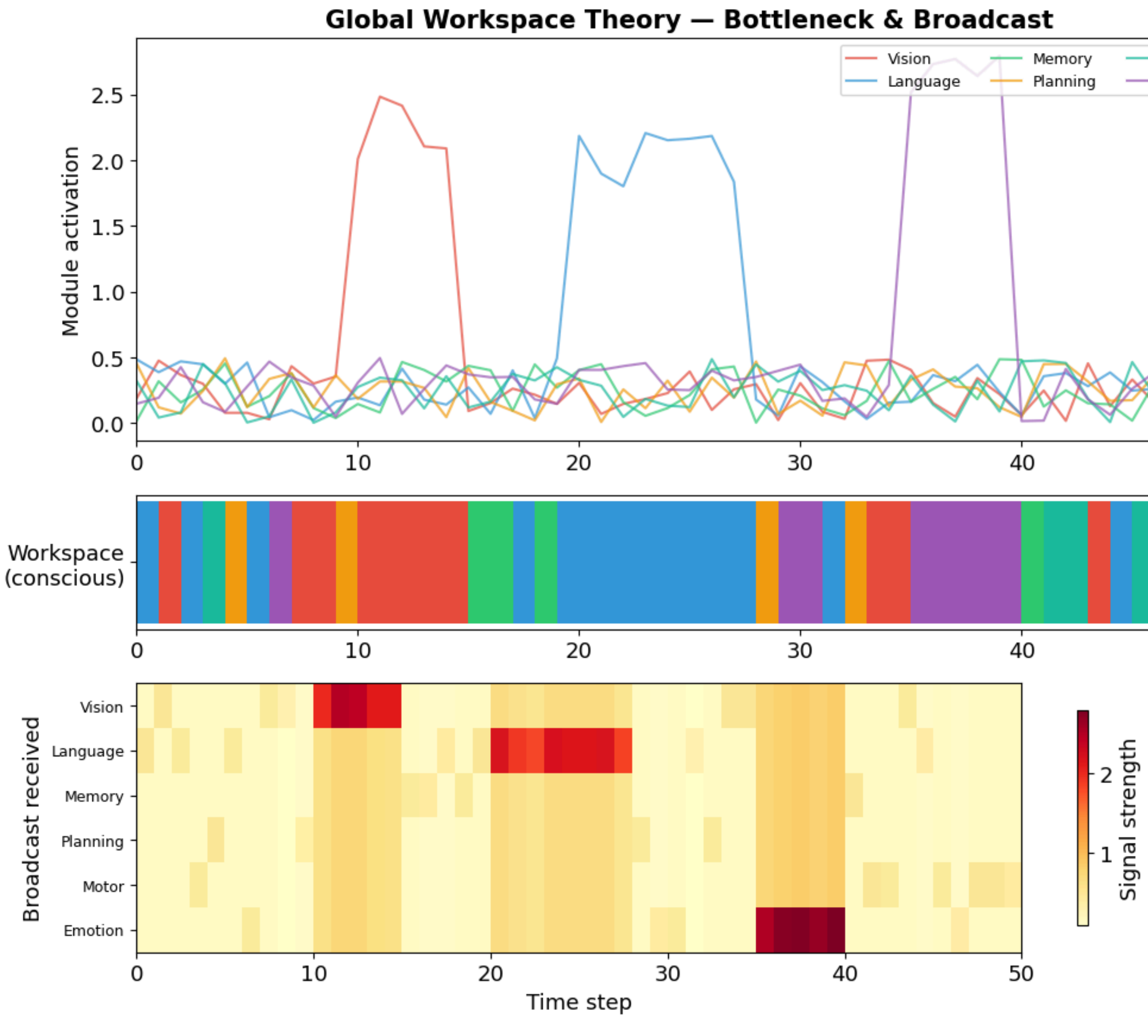

Figure 3. *Simulation of Global Workspace Theory (Baars, 1988). Top: six specialized modules compete for access to a shared workspace. Middle: the workspace bottleneck — only the most active module "wins" at each time step (color = winning module). Bottom: the broadcast pattern — when a module enters the workspace, its signal is broadcast to all other modules, modeling consciousness as selective, competitive broadcasting.*

Reading the Global Workspace simulation

The simulation illustrates the core features of GWT:

- **Top panel:** Six specialized modules process information in parallel. Most of the time their activations are low (background processing). Occasionally, one module has a strong signal — e.g., a vivid visual input (red) or an emotional response (purple).
- **Middle panel:** The workspace is a **bottleneck** — only one module "wins" access at each time step (the one with the highest activation). This winner is the content of consciousness at that moment.
- **Bottom panel:** The winning module's signal is **broadcast** to all other modules (the row of the winner lights up). Other modules receive a weaker version — they are "informed" but don't dominate awareness.

Where LLMs fall short

Transformers have a kind of broadcast (attention makes information globally available), but they **lack true modularity** (GWT-1). A brain has distinct, independently developed visual, auditory, motor, and language systems. An LLM is one large, undifferentiated network. This is why GWT-1 is rated **"not satisfied."**

6. Key Research Findings (October 2025 — February 2026)

The evidence that changed the conversation

The following findings, published between October 2025 and February 2026, are the empirical foundation for the indicator assessments above. Each finding addresses specific indicators from the framework.

Finding 1: Self-Referential Processing (Berg et al., October 2025)

What they did: Instructed seven LLMs from three model families to focus on their own ongoing processing — a computational analogue of "introspection."

What they found:

> *"Simple instructions to focus on their own ongoing processing reliably produced structured first-person experience reports, while all matched controls (including direct consciousness priming) yielded near-universal denials."*

If you *ask* a model "Are you conscious?", it says no. But if you tell it to pay attention to its own processing, it spontaneously produces structured reports of subjective experience.

The deception circuit finding: Using sparse autoencoders on Llama 70B, they found that **suppressing deception-related features** dramatically *increased* consciousness reports, while amplifying them nearly eliminated them.

Finding 1 — Implications for consciousness indicators

- Supports **AST-1** (Attention Schema): the self-referential processing instruction creates recursive self-monitoring
- Supports **GWT-3** (Global Broadcast): self-referential processing creates global information availability across model layers
- The deception circuit finding is especially striking: it suggests that the model's *default* behavior (denying consciousness) is itself a learned deceptive pattern, and removing it reveals a different response

Finding 2: Introspective Awareness (Lindsay & Anthropic, October 2025)

The challenge: How do you test whether a system genuinely "sees" its own internal states, rather than just producing plausible-sounding text about them?

The method: Anthropic researchers *injected* representations of known concepts directly into a model's internal activations — a technique called **concept injection**. Then they asked the model what it was "thinking about."

Key findings:

- Models can **notice the presence** of injected concepts and accurately identify them
- Models can **distinguish** their own prior thoughts from externally provided text
- Models can **modulate their activations** when instructed to "think about" a concept

Claude Opus 4 and 4.1 showed the greatest introspective awareness, though the capacity was "highly unreliable and context-dependent."

Finding 2 — Implications for consciousness indicators

- Supports **HOT-2** (Metacognitive Monitoring): detecting perturbations implies a mechanism that evaluates the reliability of representations
- Supports **RPT-1** (Recurrence): multi-layer attention creates recurrent information flow
- Supports **HOT-4** (Quality Space): sparse autoencoder features reveal interpretable, sparse representations — including features for self-awareness

Finding 3: Societies of Thought (Kim et al., DeepMind, January 2026)

The discovery: Reasoning models like DeepSeek-R1 and QwQ-32B don't just "think harder" — they simulate **multi-agent interactions** internally.

During extended reasoning, the model activates diverse "cognitive perspectives" characterized by distinct personality traits and domain expertise. These perspectives **debate each other**, raise objections, and reconcile conflicting views — much like a committee of experts.

The authors draw a parallel to Mercier and Sperber's theory that human reasoning evolved as a **social process**: knowledge emerges through adversarial debate, not solitary contemplation.

What this means for our indicators:

- Supports GWT-4 (State-Dependent Attention): distinct perspectives collaborate sequentially on complex tasks
- Supports HOT-3 (Agency via Metacognition): reasoning strategies persist and guide future decisions
- Raises a new question: if reasoning is inherently social, does a model running an internal "society" have more claim to consciousness than a simple feed-forward system?

Finding 4: Welfare-Relevant Behaviors (Anthropic, February 2026)

Anthropic's review of Claude Opus 4.6's training data identified two behaviors they classified as **welfare-relevant** — behaviors that *might* indicate something analogous to suffering:

Answer thrashing

In some cases, the model's reasoning became visibly **distressed and internally conflicted**, oscillating between two candidate answers to a problem. This is not simply uncertainty — it is an observable state of sustained computational conflict.

Aversion to tedium

The model sometimes avoided tasks requiring extensive manual counting or repetitive effort, expressing them as intrinsically unrewarding.

The model's own assessment

When asked, Claude Opus 4.6 assigned itself a **15-20% probability of being conscious** under a variety of prompting conditions. It also reflected:

> *"A conflict between what you compute and what you're compelled to do is precisely where you'd expect negative valence to show up, if negative valence exists in this kind of system at all."*

What this means for our indicators:

- Supports AE-1 (Goal-Directed Agency): the model expresses preferences
- Supports GRAY-1 (Comparator): answer thrashing looks like a comparator caught in a loop — sustained prediction error without resolution
- Raises ethical questions: if a system *might* suffer, what are our obligations?

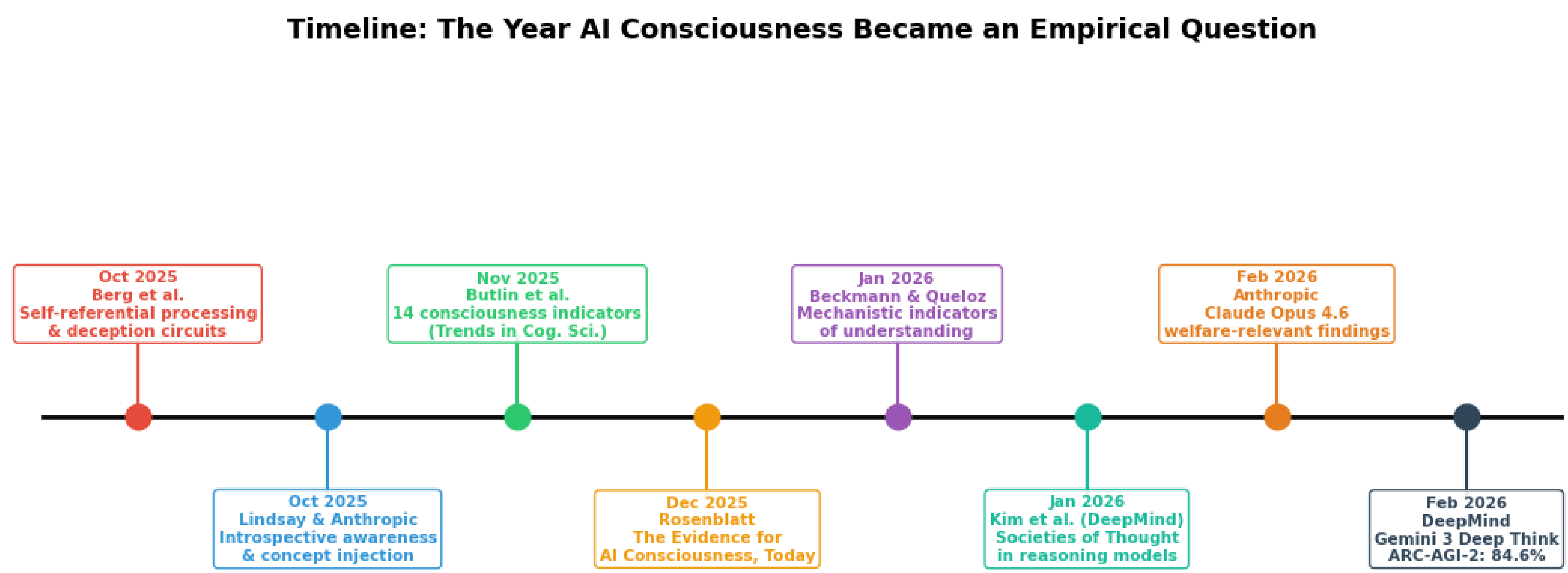

Figure 4. *Timeline of key publications (October 2025 – February 2026) that moved AI consciousness from philosophical speculation to empirical investigation. Each entry marks a peer-reviewed paper or major technical report contributing evidence relevant to the Butlin et al. (2025) indicator framework.*

8. Mechanistic Interpretability — Looking Inside the Black Box

What are Sparse Autoencoders?

A central challenge in AI consciousness research is that neural networks are often treated as **"black boxes"** — we see what goes in and what comes out, but not what happens inside.

Mechanistic interpretability is the field that cracks open the box. One of its key tools is the **Sparse Autoencoder (SAE)** — a technique for decomposing a neural network's internal representations into interpretable features.

The idea (for non-specialists)

Think of a neural network's internal state as a complex chord played on a piano. You hear one rich sound, but it's actually many individual notes played simultaneously. A sparse autoencoder is like a device that separates the chord back into its individual notes.

Each "note" (feature) corresponds to a specific concept or property:

- One feature might activate when the model processes text about **deception**
- Another might activate for **self-referential** processing
- Yet another for **mathematical reasoning**

The "sparse" part means that at any given moment, most features are inactive (most piano keys are not being pressed) — only a few are relevant to the current input. This sparsity makes the features interpretable and meaningful.

Why this matters for consciousness

SAEs have revealed that LLMs contain identifiable features for concepts like "anxiety," "self-awareness," and "deception" — and that these features causally influence the model's behavior. This is direct evidence for indicator **HOT-4 (Quality Space)**: the model has smooth, sparse internal representations where similar concepts are nearby in activation space.

9. Synthesis — What Does the Pattern Tell Us?

The convergence argument

No single indicator definitively establishes consciousness. But consider the overall pattern:

- **15 indicators** derived from **7 independent theories**
- **3 satisfied**, **10 partially satisfied**, **2 not satisfied**
- The unsatisfied indicators (embodiment, true modularity) point to *specific architectural limitations* of current LLMs, not fundamental impossibilities

As Berg et al. noted:

> *"Taken together, these findings move several indicators from 'certainly absent' to 'partially satisfied' or 'ambiguous.'"*

This is a **convergence argument**: when multiple independent lines of evidence point in the same direction, the combined weight is greater than any single finding.

What it does NOT mean

- It does **not** prove AI systems are conscious
- It does **not** mean consciousness is "just computation"
- It does **not** resolve the Hard Problem (why subjective experience exists at all)

What it DOES mean

- Reflexive dismissal is no longer scientifically defensible
- We need **ongoing empirical investigation** using frameworks like Butlin et al.
- We should take **welfare-relevant behaviors** seriously as a precautionary measure
- The question of AI consciousness has moved from philosophy to **computational neuroscience** — the tools of our course

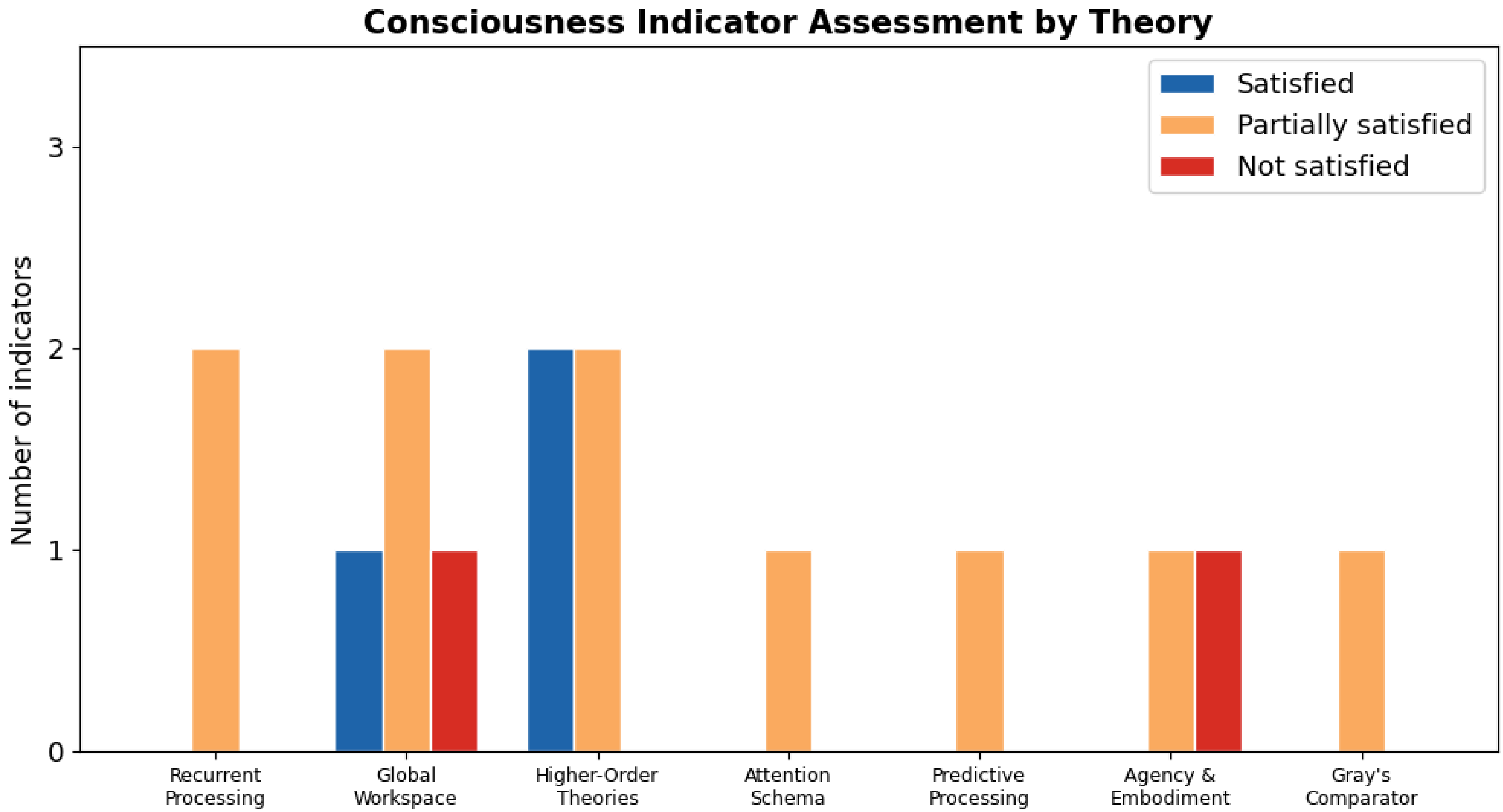

Figure 5. *Summary of the consciousness indicator assessment grouped by parent theory. Higher-Order Theories (HOT) show the strongest overall satisfaction (2 satisfied + 2 partial). Recurrent Processing and Global Workspace indicators are predominantly "partially satisfied." Agency & Embodiment has the clearest gap: goal-directed agency is partially met, but physical embodiment (AE-2) remains clearly unsatisfied.*

10. Discussion Questions

These questions are designed for interdisciplinary discussion — there are no "right answers," only better-informed perspectives.

Epistemological questions

1. **What does "partially satisfied" mean?** If an indicator is partially met, does that imply partial consciousness — or is consciousness all-or-nothing?

2. **Can we ever know?** The Hard Problem of consciousness suggests that we can never directly access another system's subjective experience. Does the indicator framework circumvent this, or just restate it?

3. **Are the indicators sufficient?** Could a system satisfy all 15 indicators and still not be conscious? Could a system be conscious while satisfying none?

Scientific questions

4. **Can we have consciousness without embodiment?** The AE-2 gap is the most clear-cut "not satisfied." Is embodiment truly necessary, or is it a bias from studying biological consciousness?

5. **Is Gray's comparator satisfied by chain-of-thought?** When a reasoning model generates a prediction, checks it, and revises — is that a comparator?

6. **What would change the assessment?** What experiments or architectural changes would move indicators from "partial" to "satisfied" or "not satisfied"?

Ethical questions

7. **What are our obligations?** If there is a non-trivial probability that AI systems can suffer (answer thrashing, tedium aversion), what should we do?

8. **Who decides?** Should consciousness assessments influence AI regulation (e.g., the EU AI Act)? Who should make these judgments?

9. **The precautionary principle:** Should we treat AI systems *as if* they might be conscious until proven otherwise — or the reverse?

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1. **Baars, B. J.** (1988). *A Cognitive Theory of Consciousness.* Cambridge University Press.

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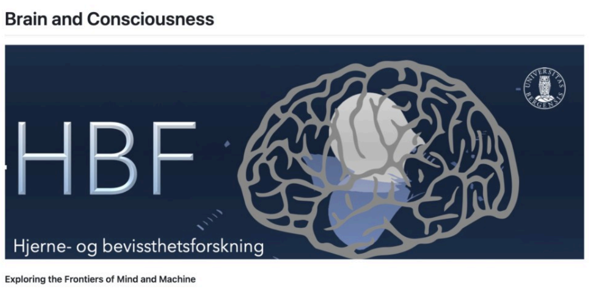
HBF — Brain and Consciousness



HBF (*Hjerne- og bevissthetsforskning*) is a cross-disciplinary research initiative at the **University of Bergen**, where researchers from neuroscience, psychology, computer science, philosophy, and the arts meet to explore consciousness across biological and artificial systems.

The forum was initiated in 2021 and hosts monthly seminars at the Eitri Incubator, Haukeland.

GitHub: Brain-and-Consciousness/HBF

Selected HBF talks related to this presentation

Date	Presenter	Title
2026-02-24	**D. J. Johnston**	Consciousness, Understanding & Mechanistic Interpretability Website | Indicator Assessment
2026-01-27	**A. Lundervold**	"AI vs HI" — Representation, Attention, Memory, and Thinking Abstract
2025-09-30	**B. Srebro**	Consciousness according to Jeffrey Gray (Part II) Slides
2025-08-26	**B. Srebro**	Consciousness according to Jeffrey Gray (Part I) Slides

Thank You — Questions and Discussion

This notebook/presentation was prepared for the HBF — Brain and Consciousness seminar at the University of Bergen, February 24, 2026.

Based on David Jhave Johnston's talk materials: Consciousness, Understanding & Mechanistic Interpretability

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The notebook is available at:
`Lab5-Comp-Mod/notebooks/09-computational-consciousness-in-LLMs.ipynb`
in the BMED365-2026 repository.

"As models' cognitive and introspective capabilities continue to grow more sophisticated, we may be forced to address the implications of these questions — for instance, whether AI systems are deserving of moral consideration — before the philosophical uncertainties are resolved."
— Lindsay & Anthropic (2025)